

Article

Adsorption of congo red on mesoporous activated carbon prepared by CO₂ physical activation



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ABSTRACT

This research demonstrates the production of mesoporous activated carbon from sargassum fusiforme via physical activation with carbon dioxide. Central composite design was applied to conduct the experiments at different levels by altering three operating parameters. Activation temperature (766–934 °C), CO₂ flow rate (0.8–2.8 L·min⁻¹) and activation time (5–55 min) were the variables examined in this study. The effect of parameters on the specific surface area, total pore volume and burn-out rate of activated carbon was studied, and the influential parameters of methylene blue adsorption value were identified employing analysis of variance. The optimum conditions for maximum methylene blue adsorption value were: activation temperature = 900 °C, activation time = 29.05 min and CO₂ flow rate = 1.8 L·min⁻¹. The activated carbon produced under optimum conditions was characterized by BET, FTIR and SEM. The adsorption behavior on congo red was studied. The effect of parameters on the adsorbent dosage, temperature, PH and initial congo red concentration was investigated. The adsorption properties of the activated carbon were investigated by kinetics. The equilibrium removal rate and maximum adsorption capacity reaches up to 94.72%, 234 mg·g⁻¹, respectively when initial congo red concentration is 200 mg·L⁻¹ under adsorbent dosage (0.8 g·L⁻¹), temperature (30 °C), PH7.

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1. Introduction

A significant number of pollutants are daily released into rivers, sea water and soils with developments of industrialization and urbanization [1]. Various organic compounds such as pesticides, dioxins, chlorophenols and dyes are found in the contaminated water [2–4]. Some organics can induce skin allergy and mutagenesis, leading to cancer and various diseases [5]. Moreover, due to their composite molecular structures, these contaminants are reported as recalcitrant and non-biodegradable molecules, which block the penetration of solar light and, consequently, hinder the water oxygenation. In addition, most dyes have complex and very stable aromatic structures, and thus tend to accumulate in nature.

Nowadays, several physical, chemical and biological treatments have been reported for dye remediation [6–9]. Among the proposed treatment methods, adsorption technologies are regarded as one of the competitive methods due to potential low-cost, high efficiency, simplicity of design, and operation [10]. It is very important to select a suitable adsorbent in water treatment to obtain the best performance for the removal of water pollutants [11]. Activated carbon is one of the most commonly used solid adsorbents [12,13], which offers high adsorption performance, well-developed porous structure and large specific surface area. With the shortage of petroleum coke, coal and other nonrenewable resources, along with serious damage to forestry, the raw materials for producing activated carbon are becoming scarce. Therefore, developing new kinds of renewable resources to prepare activated carbon is an inevitable trend of sustainable development. At present, the preparation of carbon materials from large algae has become a research hotspot [14], which relieves the problem of seawater eutrophication and heavy metal pollution. The raw materials commonly include oceanica fibers [14], sargassum horneri [15], brown algae [16] and sargassum fusiforme. Sargassum fusiforme is the warm temperate algae specific to the western north pacific, widely distributed along

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the coast of China, Japan and north Korea [17,18]. Sargassum fusiforme is hypertrophic and nutritious, and the planting area of sargassum fusiforme is over 1000 ha in the Dongtou county of Zhejiang province. It is said that the annual production of sargassum fusiforme is approximately 8000 tons. During the production process of sargassum fusiforme, large amount of scraps were obtained. However, at present, the investigation on reuse of these scraps has not been profound enough, calling for the development of effective utilization methods.

The common preparation methods of activated carbon include metal catalytic activation method [19,20], chemical method [21–23], template method [24,25] and physical method. In recent years, physical activation method has been attracted more and more attention because of its advantages of low preparation cost, simple operation, little environmental pollution, low equipment requirement, etc. Physical activation process takes place in two stages: carbonization stage and activation stage. During the carbonization stage, the raw material is pyrolyzed in an inert atmosphere at a medium–high temperature (300–700 °C). Throughout this process, the breaking of the less stable bonds releases the volatile fraction of the raw material, which is formed by permanent gases and tars. Then a carbonaceous residue enriched in carbon aromatic rings is obtained, the so-called char, which has a basic porous structure. During the activation stage, char is activated at a higher temperature (700–1000 °C) in the presence of an activating agent. During this process, the carbonaceous matrix of the fuel is exposed to a reducing atmosphere and undergoes several heterogeneous reforming reactions leading to a partial gasification of the char, developing a large porous structure and increasing its specific surface area.

The chars produced by the carbonization process react with oxidizing gases such as steam (H₂O), carbon dioxide (CO₂), air or any mixture of these gases, resulting in the evolution of carbon oxides from the carbon surface. Generally, the use of carbon dioxide is preferred due to its lower reactivity at high temperature, which makes the activation process easier to control.

In this work, we prepared mesoporous activated carbon through CO₂ physical activation method from the scraps of sargassum fusiforme. Then mesoporous activated carbon was employed for removal of congo red, an example of diazo dyes prepared by coupling tetrazotised benzidine with two molecules of naphthionic acid, in aqueous solution. This investigation not only focused on the resource utilization of the remaining sargassum fusiforme scraps, but also provided a new idea for the treatment of dye wastewater.

2. Materials and Methods

2.1. Materials and instruments

Sargassum fusiforme scraps were purchased from Tongtong county, Zhejiang province. Main chemical reagents and instruments are listed in Table 1a and Table 1b.

2.2. Preparation of activated carbon

The preparation experimental device sketch map was shown in Fig. 1. Sargassum fusiforme scraps were washed and then dried for 12 h in the drying oven. After shattered for 60 s, scraps were 550 °C carbonized in muffle furnace to obtain biochar. Then biochar was

Table 1b
Main instruments.

Instruments	Brand	Manufacturer
Drying oven	DHG-9070C	Hangzhou Lantian Instrument Co. Ltd.
Spectrophotometer	Lambda 35	PerkinElmer
Vacuum tube furnace	SK3-3-12-8	Hangzhou Zhuochi Instrument Co. Ltd.
Specific surface and aperture analyzer	3H-2000PS1	Beijing Beishide Instrument Co. Ltd.
Fourier transform infrared spectrometer	Tensor II	German Bruker Company
Scanning electron microscope	HitachiS-4700 (II)	Japanese Rili Company
Reciprocating oscillator	SHA-B	Changzhou Aohua Instrument Co. Ltd.

taken into the vacuum tube furnace for activation. The activation process was divided into three parts: firstly, vacuum tube furnace was programmed to rise up to a predetermined activation temperature (750–950 °C) at a rate of 10 °C·min⁻¹. Secondly, under the condition of constant activation temperature, carbon dioxide was inserted at a constant flow rate (0.6–2.4 L·min⁻¹) for a certain period of time (15–60 min). Thirdly, the system was cooled to room temperature. It is worth mentioned that in the whole activation process, nitrogen was inserted into the vacuum tube furnace at a flow rate of 0.5 L·min⁻¹ to maintain the activation pressure of the system. 1 mol·L⁻¹ HCl solution was employed to boil the activation sample for 30 min, then the sample was washed for three times by boiled HCl solution. After washed by boiled HCl solution, the sample was filter washed to neutral by deionized water (95 °C) and the filter cake was dried for 24 h in 105 °C drying oven to obtain activated carbon.

2.3. Response surface central composite method

To determine the proper activation temperature, activation time, and CO₂ flow rate, response surface central composite method was carried out using methylene blue as model adsorbate.

In order to investigate the effect of activation temperature, activation time and CO₂ flow rate on specific surface area, pore volume, burnout rate and methylene blue adsorption capacity of activated carbon, single-factor experiments were carried out at different activation temperature (750, 800, 850, 900, 950 °C, activation time: 30 min, CO₂ flow rate: 1.8 L·min⁻¹), activation time (0, 15, 30, 45, 60 min, activation temperature: 900 °C, CO₂ flow rate: 1.8 L·min⁻¹), CO₂ flow rate (0, 0.6, 1.2, 1.8, 2.4 L·min⁻¹, activation temperature: 900 °C, activation time: 30 min).

Based on the results of single-factor experiments, central composite design (response: methylene blue adsorption capacity; independent variable: activation temperature, activation time and CO₂ flow rate) was made by employing Design Expert 8.0.6 software. Factors and levels are listed in Table 2.

Activated carbon specific surface area and pore volume were measured by specific surface and aperture analyzer. Burnout rate (Y₁) was calculated using the following equation:

$$Y_1 = \frac{m_0 - m_1}{m_0} \quad (1)$$

where m_0, m_1 is activated carbon mass before and after activation process.

The adsorption performance of methylene blue (Y₂) was tested by following steps: activated carbon (0.1 g, 190–240 mesh) was mixed with certain volume of methylene blue solution (1.5 g·L⁻¹) in a conical flask (100 ml), then mixer was oscillated for 20 min by reciprocating oscillator and filtered by medium speed qualitative filter paper (diameter: 12.5 cm). Compared with absorbance of standard CuSO₄

Table 1a
Main reagents.

Reagents	Mass fraction	Manufacturer
Hydrochloric acid	0.37	Hangzhou Chemical Reagent Co. Ltd.
Sodium hydroxide	0.96	Xilong Science Co. Ltd.
Congo red	0.98	Aladin Reagent Co. Ltd.
Methylene blue	0.70	Aladin Reagent Co. Ltd.

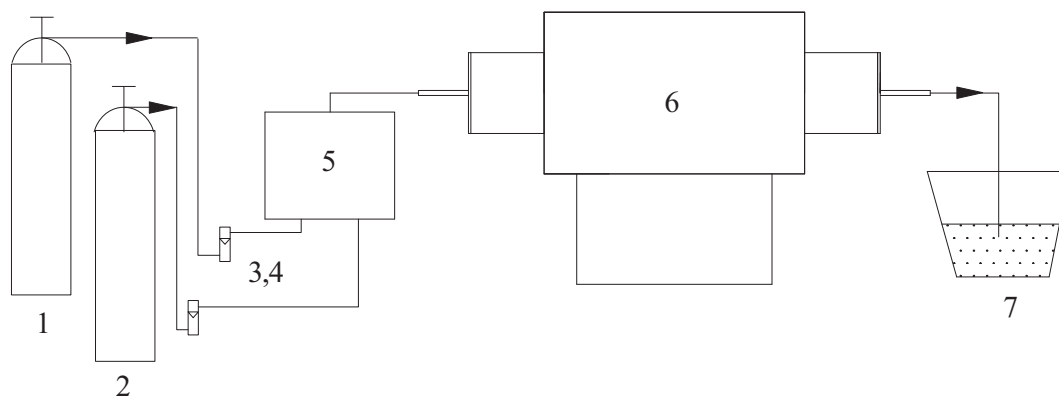


Fig. 1. The preparation experimental device sketch map of activated carbon. 1. carbon dioxide cylinder; 2. nitrogen cylinder; 3,4. rotor flow meter; 5. buffer tank; 6. vacuum tube furnace; 7. tail gas adsorber.

filter solution, the absorbance of filtrate was measured by spectrophotometer at a wavelength of 665 nm to determine methylene blue adsorption capacity.

2.4. Characteristic of adsorbent

Nitrogen adsorption–desorption isotherms at $-196\text{ }^{\circ}\text{C}$ were obtained using a gas adsorption analyzer. A sample of 0.05 g was degassed at $200\text{ }^{\circ}\text{C}$ for $>3\text{ h}$ prior to isotherm measurement. The specific surface area was based on the Brunauer–Emmett–Teller (BET) theory [26] and was calculated using the volume of N_2 adsorbed at relative pressures of 0.05–0.1. The liquid N_2 volume in relation to the N_2 volume adsorbed at a relative pressure of 0.999 was determined as the total pore volume. The pore size distribution was evaluated using density functional theory (DFT) [27,28]. The surface characteristics of adsorbent were observed with a scanning electron microscope. Functional groups of activated carbon were analyzed by the Fourier transform infrared spectrometer (resolution: 4 cm^{-1} ; test range: $400\text{--}4000\text{ cm}^{-1}$).

2.5. Adsorption test

Adsorption test was carried out in the oscillator at a rotate speed of 150 rpm. Activated carbon (0.01 g, 0.02 g, 0.04 g, 0.06 g, 0.08 g, 0.10 g) was mixed with CR solution (25 ml; concentration: 50, 100, 150, 200 $\text{mg}\cdot\text{L}^{-1}$) under the condition of temperature (25, 30, 35, 40, $45\text{ }^{\circ}\text{C}$), PH value (2, 4, 5, 7, 9, 11) for a certain period of time (5, 10, 20, 30, 50, 90, 120, 180 min). Adsorbent was input into a 100 ml conical flask containing 25 ml of CR solution, and the adsorption test was carried out in the oscillator at a rotate speed of $150\text{ r}\cdot\text{min}^{-1}$.

2.6. Determination of adsorption capacity and removal rate

Adsorption capacity and removal rate are determined by initial CR concentration (c_0), final CR concentration (c_t), activated carbon mass (m) and CR volume (V). The equations can be given as:

$$R = \frac{c_0 - c_t}{c_0} \quad (2)$$

Table 2
Factors and levels of the central composite design

Factors	Numbers	Levels				
		1.68	1	0	−1	−1.68
Activation temperature	A/ $^{\circ}\text{C}$	934	900	850	800	766
Activation time	B/min	55	45	30	15	5
CO_2 flow rate	C/ $\text{L}\cdot\text{min}^{-1}$	2.8	2.4	1.8	1.2	0.8

$$Q = \frac{(c_0 - c_t)V}{m} \quad (3)$$

To determine c_t , the absorbance (A) of CR solution after adsorption process was measured by spectrophotometer at optimum absorption wavelength (499 nm). Absorbance and concentration are in accordance with Lange Beer's law. Standard curve of CR solution is shown in Fig. 2.

2.7. Kinetic study

The pseudo first order equation is described as follows [29]:

$$\lg q_e - q_t = \lg q_e - \frac{k_1 t}{2.303} \quad (4)$$

where q_t and q_e ($\text{mg}\cdot\text{g}^{-1}$) are the adsorption capacities at time t and equilibrium, respectively, and k_1 (min^{-1}) is the pseudo-first-order rate constant.

The pseudo second order equation is as follows:

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (5)$$

where k_2 ($\text{g}\cdot(\text{mg}\cdot\text{min})^{-1}$) is the pseudo-second-order rate constant.

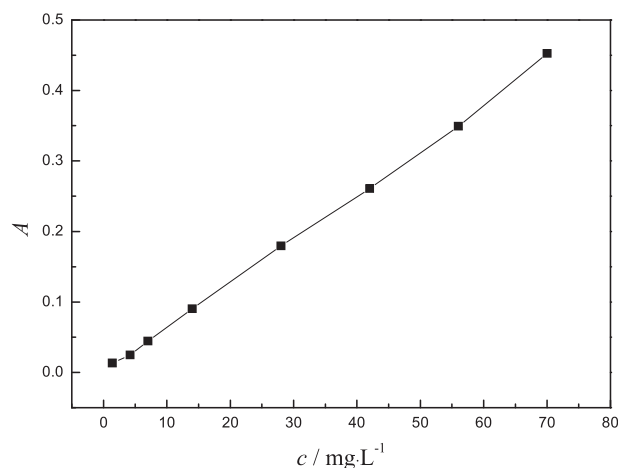


Fig. 2. Standard curve of CR solution.

3. Results and Discussion

3.1. Single factor experiment results

Fig. 3 shows the effect of activation temperature on burnout rate, specific surface area and pore volume. Under low temperature, due to insufficient activation process, the porous structure is not developed enough, thus leading to small specific surface area and pore volume. When temperature reaches 900 °C, specific surface area and pore volume can reach peak value. At much higher temperature, hole wall collapses and pore structure is destroyed, which is harmful for adsorption process.

From Fig. 3, we can see that under low temperature, low CO₂ flow rate and short activation time, due to insufficient activation process, the porous structure is not developed enough, thus leading to small specific surface area and pore volume. In the opposite case, activation process is overly adequate, leading to hole wall collapsing and pore structure is destroyed, which is harmful for adsorption process. No matter what factors, the monotonically increasing tendency of burnout rate is very clear. In central composite design (CCD), we select the following factor value range (activation temperature: 800–900 °C; activation time: 15 min–45 min; CO₂ flow rate: 1.2 L·min^{−1}–2.4 L·min^{−1}) to continue the following analysis.

3.2. Central composite design

CCD experimental results are listed in Table 3. Three-dimensional regression fit was performed on methylene blue adsorption data:

$$Y_2 = 236.53 + 8.03A + 15.67B + 2.14C - 32.39AB - 6.86AC - 18.26BC - 13.93A^2 - 43.05B^2 - 40.87A^2B - 16.20A^2C \quad (7)$$

An analysis of variance was implemented to find those process parameters along with their confidence level and interactions based on degree of freedom, sums of squares, *F*-test and *P* value. The results are detailed in Table 4. Among these three parameters, activation time is the most contributing parameter, and has the greatest influence ($F(B) > F(A) > F(C)$, $F(B^2) > F(AB) > F(A^2) > F(BC) > F(AC) > F(C^2)$). CO₂ flow rate has minimal impact on response value. *P* value indicates the significance of a parameter or a model. We find that *P* value of the model is lower than 0.0001, showing that Eq. (4) fits the experimental data very well.

To make sure whether this equation can accurately reflect the fact, linear fit graph (Fig. 4) between predicted and experimental values was used to calculate the correlation coefficient (*R*²), which value is 0.9743, indicating that Eq. (4) can predict the experimental value properly.

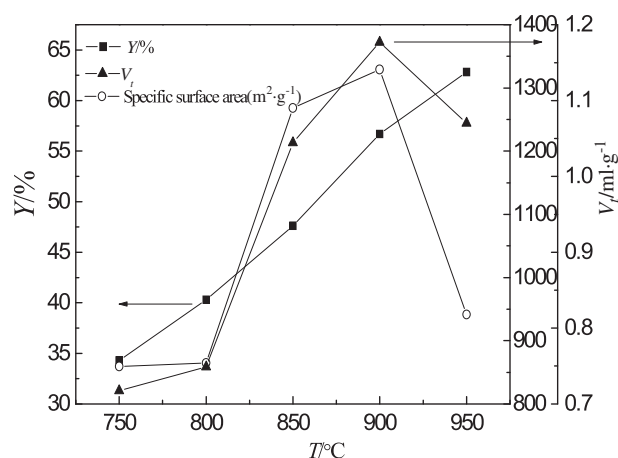
To maximize methylene blue adsorption capacity, the operating parameters were optimized adopting Design Expert 8.0.6 software, and the following optimal conditions were obtained:

- Activation temperature: 900 °C
- Activation time: 29.05 min
- CO₂ flow rate: 1.8 L·min^{−1}

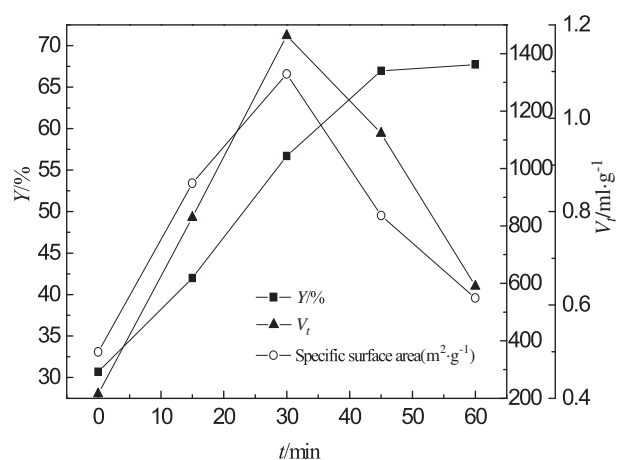
The following adsorbent was obtained under the condition of 900 °C, 30 min and 1.8 L·min^{−1}.

3.3. Characteristic of adsorbent

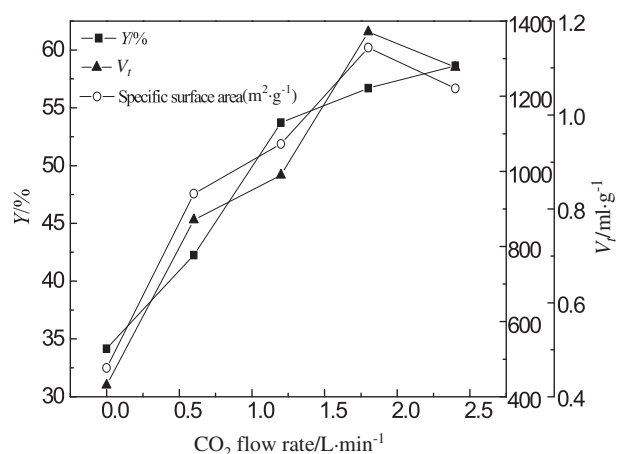
Activated carbon prepared from optimal condition was characterized by BET method and the meso-porosity is up to 65%. The N₂ adsorption isotherm and pore size distribution curves of activated carbon obtained in the experiment are shown in Fig. 5. As can be seen



(a) Activation temperature



(b) Activation time



(c) CO₂ flow rate

Fig. 3. Effect of different factors on burnout rate, specific surface area and pore volume *V_t*.

from N₂ adsorption/desorption isotherm, a gradual increase in isotherm slope at relative pressures > 0.1 and a hysteresis loop at a high relative pressure, indicating the existence of mesopores, this phenomenon reveals that the isotherm conforms to the standard type IV. In the low relative pressure zone, the adsorption isotherm shows a rapid increase

Table 3
Central Composite Design and experimental results.

Numbers	A/°C	B/min	C/L·min ⁻¹	Y ₂ /mg·g ⁻¹
1	800	15	2.4	165.65
2	766	30	1.8	194.22
3	900	15	1.2	178.57
4	850	30	0.8	231.12
5	850	5	1.8	97.60
6	850	30	1.8	239.00
7	900	15	2.4	239.28
8	850	30	2.81	238.33
9	800	15	1.2	148.81
10	900	45	1.2	104.79
11	850	30	1.8	245.80
12	850	30	1.8	233.48
13	800	45	2.4	148.81
14	850	55	1.8	150.30
15	900	45	2.4	82.25
16	800	45	1.2	194.42
17	934	30	1.8	218.37
18	850	30	1.8	239.76
19	850	30	1.8	225.68
20	850	30	1.8	254.25

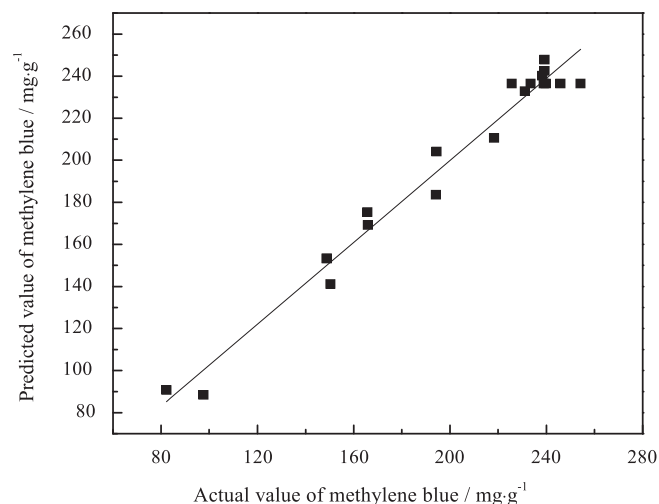
in the adsorption volume, which is caused by the microporous structure of activated carbon. When the adsorption relative pressure is about 0.4, obvious hysteresis ring appears, indicating that the activated carbon has abundant mesopores. According to Fig. 5(b), the values of pore diameter are concentrated around 4 nm, in the mesoporous range (2–50 nm).

The external characteristic of the adsorbent can be clearly observed by SEM. As can be seen from Fig. 6(a), sargassum fusiforme is a massive crystal with almost no pores. After carbonization (Fig. 6(b)), a series of pores appeared and the structure of raw material started to be destroyed under high carbonization temperature (550 °C). During activation process, the adsorbent surface morphology continues to be irregular and porous (Fig. 6(c)), and the pores are in good condition, which provides a large number of activated sites for adsorbate to enter in.

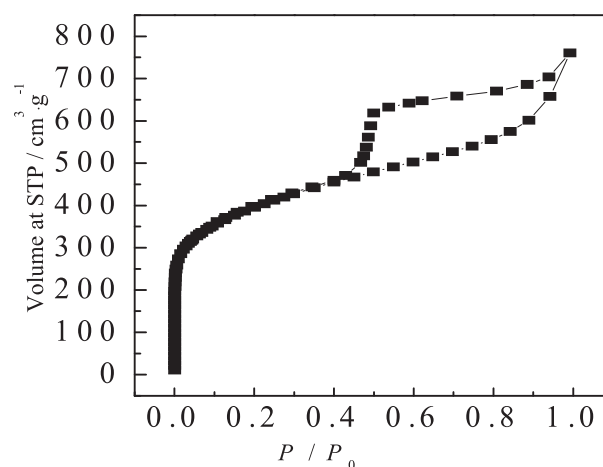
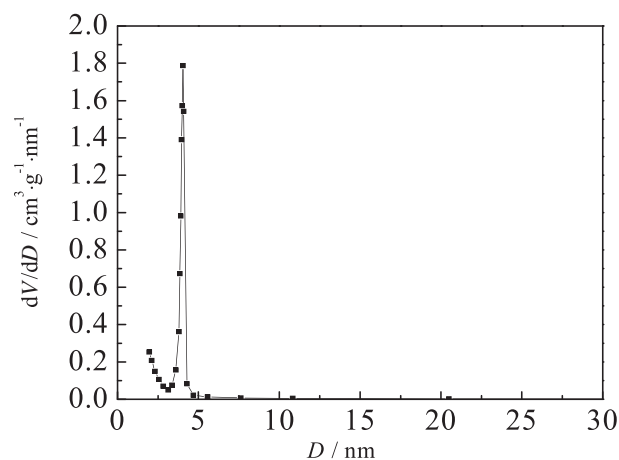
The FT-IR spectra of biochar and activated carbon are shown in Fig. 7. It can be seen that activated carbon exhibits a broad band at 3420 cm⁻¹ due to -OH stretching vibration. The broad absorption band at 1124 cm⁻¹ is assigned to C-O stretching vibration. And, the band among 1650–1700 cm⁻¹ on activated carbon is caused by aromatic (carbonyl/carboxyl groups) (C=O). From the spectra of the biochar, it can be clearly observed the peaks mainly due to organic structures in the raw material. The bands at 2930 cm⁻¹, 1431 cm⁻¹ and 878 cm⁻¹ are caused by aliphatic C-H stretching vibration, bending vibration and out plane bending vibration, respectively. Aromatic bonds (O-H) or out plane bending can be observed at 620 cm⁻¹ in biochar. It can be concluded that H element was removed in bulk throughout the

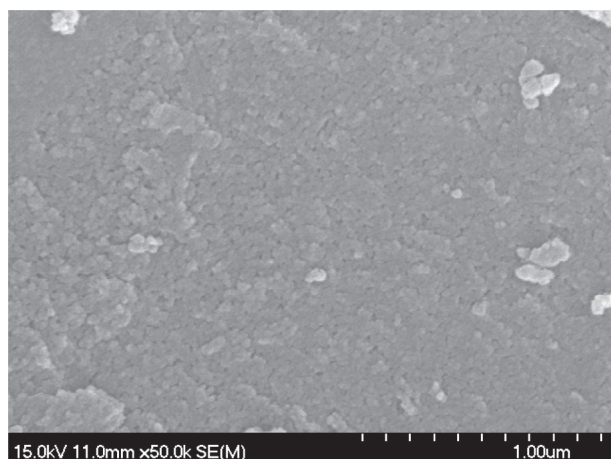
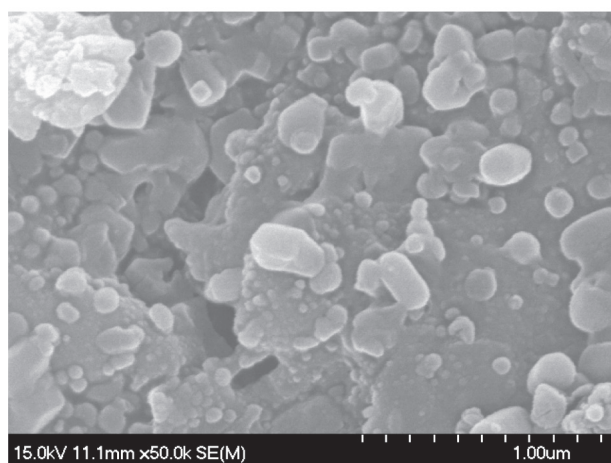
Table 4
Significance of regression Eq. (4).

Items	Sum of square	Degrees of freedom	F-ratio	P
Model	48,850.23	10	34.14	<0.0001
A	880.46	1	6.15	0.0350
B	1388.65	1	9.71	0.0124
C	25.99	1	0.18	0.6800
AB	8395.49	1	58.68	<0.0001
AC	376.75	1	2.63	0.1391
BC	2668.88	1	18.65	0.0019
A ²	2825.99	1	19.75	0.0016
B ²	26,971.38	1	188.51	<0.0001
A ² B	5535.16	1	38.69	0.0002
A ² C	869.49	1	6.08	0.0359
Residual	1287.68	9		
Missing item	803.03	4	2.07	0.2224
Net error	484.65	5		
Total dispersion	50,137.91	19		

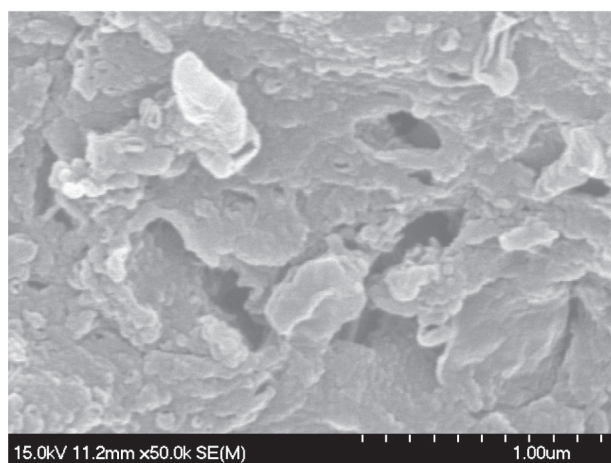
**Fig. 4.** Comparison of predicted and experimental values.

activation progress. Moreover, -COOH functional group was found on the surface of activated carbon.

**(a)** Adsorption isotherm**(b)** Pore size distribution curve**Fig. 5.** Nitrogen adsorption-desorption isotherms and pore size distribution.

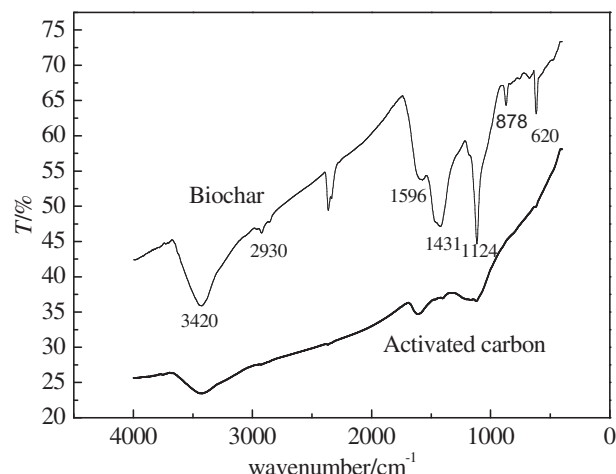
(a) *Sargassum fusiforme*

(b) Biochar



(c) Activated carbon

Fig. 6. SEM images of raw material, biochar and activated carbon.

Fig. 7. FT-IR image of *Sargassum fusiforme* based biochar, activated carbon.

of activated carbon. From Fig. 8, we can see that adsorbent dosage has a great influence on adsorption capacity. Adsorption capacity may decrease from $121.81 \text{ mg} \cdot \text{g}^{-1}$ to $21.93 \text{ mg} \cdot \text{g}^{-1}$ with an increase in activated carbon dose from 0.4 to $4 \text{ g} \cdot \text{L}^{-1}$. The lower adsorption capacity with an increasing dose of adsorbent is basically because the adsorbent's adsorption capacity is based on the adsorbent's adsorption reaction point rather than the reaction point percentage. An increased amount of adsorbent additive can result in an unsaturated adsorption point. What's more, the increase of adsorbent dosage was expected to cause the agglomeration of adsorbents, making a certain mass of adsorbents unavailable to the adsorption, resulting in the decrease of adsorption volume. The removal rate experiences a sharp increase from 49.7% to 83.8% with an increase in activated carbon dose from 0.4 to $0.8 \text{ g} \cdot \text{L}^{-1}$. When adsorbent dosage value is larger than $0.8 \text{ g} \cdot \text{L}^{-1}$, removal rate value ranges from 83% to 89% . Considering overuse of adsorbent may result in waste, $0.8 \text{ g} \cdot \text{L}^{-1}$ is the most suitable adsorbent dosage value to obtain high removal rate of congo red.

As shown in Fig. 9, the removal rate increased first and then decreased with increasing temperature, and reached the maximum removal rate at 30°C . However, the overall removal rate is higher than which under normal temperature, indicates that the adsorption process may be an endothermic reaction [30].

It can be seen from Fig. 10 that removal rate and adsorption capacity are decreasing as solution PH increases. The ionization degree and nature of congo red depends on PH values [30]. Activated carbon surface contains $-\text{COOH}$ functional group, which was changed to $-\text{COO}^-$ under

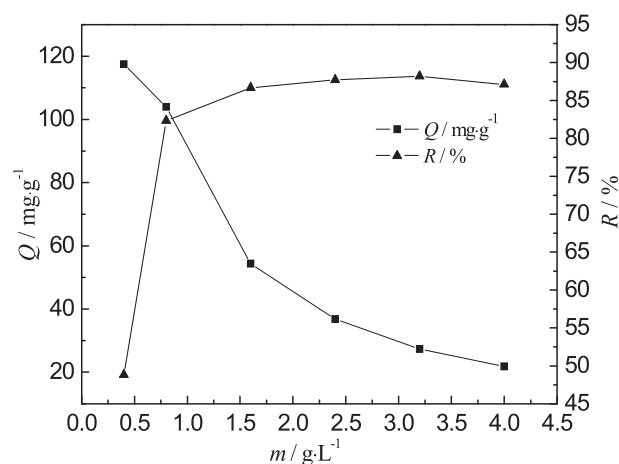


Fig. 8. Effect of adsorbent dosage.

3.4. Congo red adsorption

In this section, effect of adsorbent dosage, temperature, PH and congo red solution concentration was investigated for the application

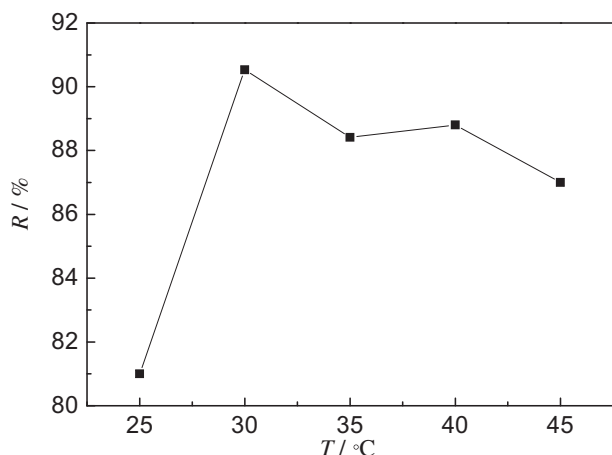


Fig. 9. Effect of temperature.

alkaline conditions, prejudice for adsorbing negatively charged dyes. Under acidic conditions, congo red existed in form of positive ion, which may replace H^+ onto carboxyl and hydroxyl, thus leading to high adsorption capacity of congo red. Removal rate and adsorption capacity reaches up to 98.27% and $123.84 \text{ mg} \cdot \text{g}^{-1}$ when solution PH is 2, and the most suitable PH value for adsorption process is 2.

Under the condition of adsorbent dosage ($0.8 \text{ g} \cdot \text{L}^{-1}$), temperature (30°C), PH (7), the adsorption process on congo red solution was investigated (Fig. 11). As congo red concentration increases, the amount of congo red adsorbed by activated carbon increases. The maximum adsorption capacity and removal rate can reach up to $234 \text{ mg} \cdot \text{g}^{-1}$, 94.7% when congo red concentration is $200 \text{ mg} \cdot \text{g}^{-1}$. The time to reach equilibrium for congo red solutions are: $50 \text{ mg} \cdot \text{L}^{-1}$ (5 min), $100 \text{ mg} \cdot \text{L}^{-1}$ (10 min), $150 \text{ mg} \cdot \text{L}^{-1}$ (30 min), $200 \text{ mg} \cdot \text{L}^{-1}$ (90 min), respectively.

Table 5 lists the saturated adsorption capacity of different adsorbents to congo red under optimal conditions. Due to proper preparation conditions, sargassum fusiforme activated carbon possesses high specific surface area and more activated sites, thus leading to high adsorption capacity to congo red.

3.5. Adsorption kinetic study

Table 6 lists the results of rate constant studies for different initial dye concentrations by the pseudo first-order and second-order models. The data shows the correlation coefficients fitted the second-order reaction rate formula ($R^2 > 0.99$), which indicates that the adsorption

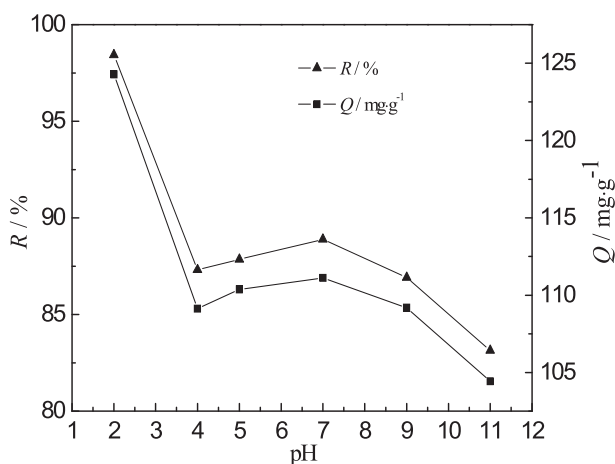
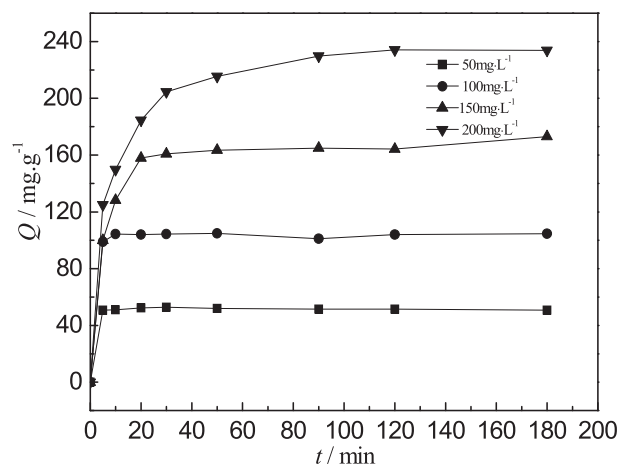
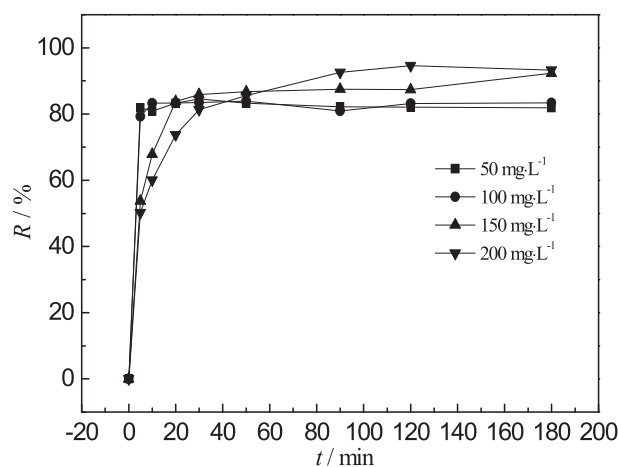


Fig. 10. Effect of PH.



(a)



(b)

Fig. 11. Effect of congo red solution concentration.

process is consistent with the second-order reaction rate theory. These suggest that the pseudo second-order adsorption mechanism is predominant and that the overall rate of the dye adsorption process appears to be controlled by the chemisorption process, in accordance with the assumption in PH study. We also observed that k_2 of congo red adsorption of the activated carbon at different solution concentrations: $50 \text{ mg} \cdot \text{L}^{-1}$ (0.06274 min^{-1}), $100 \text{ mg} \cdot \text{L}^{-1}$ (0.05842 min^{-1}), $150 \text{ mg} \cdot \text{L}^{-1}$ ($0.001718 \text{ min}^{-1}$), $200 \text{ mg} \cdot \text{L}^{-1}$ ($0.000706 \text{ min}^{-1}$) were significantly difference. This result indicated that the adsorption rate: $50 \text{ mg} \cdot \text{L}^{-1}$ solution > $100 \text{ mg} \cdot \text{L}^{-1}$ solution > $150 \text{ mg} \cdot \text{L}^{-1}$ solution > $200 \text{ mg} \cdot \text{L}^{-1}$ solution.

Table 5
Comparison between different adsorbents for congo red adsorption.

Adsorbent	Specific surface area/ $\text{m}^2 \cdot \text{g}^{-1}$	Q/ $\text{mg} \cdot \text{g}^{-1}$	References
Sargassum fusiforme activated carbon	1329	234	This study
Activated carbon fibers	2797	519	[31]
Myrtus communis activated carbon	103.5	19	[32]
Pomegranate activated carbon	40.69	10	[32]
H ₂ SO ₄ modified celery residue	24.93	238.09	[33]
ZnO-modified SiO ₂ nanospheres	34.5	83	[34]
Zeolitic imidazolate framework-67	1388	714.3	[35]

Table 6

Comparison of the first-order and second-order adsorption rate constants for different initial dye concentrations.

$c_0/\text{mg} \cdot \text{L}^{-1}$	Pseudo-first order model			Pseudo-second order model		
	k_1	q_e	R^2	k_2	q_e	R^2
50	0.09817	56.6663	0.8918	0.06274	51.2425	0.9997
100	0.07471	142.7551	0.3593	0.05842	102.1144	0.9999
150	0.08571	194.7640	0.9017	1.718×10^{-3}	161.1989	0.9979
200	0.03669	282.0762	0.9821	7.06×10^{-4}	224.7332	0.9994

4. Conclusions

In this work, sargassum fusiforme activated carbon was prepared by CO_2 activation method and used as adsorbents for congo red removal from water. Factors that affected the preparation were investigated in detail. The most contributing factor is activation time, followed by activation temperature. The least contributing factor is CO_2 flow rate. Adsorbent prepared under optimum condition possesses mesoporous structure and large specific surface area. FTIR spectra prove the existence of $-\text{COOH}$ functional group, which is beneficial for congo red adsorption. The adsorption process, factors that affected the adsorption were investigated. The results showed that the process of congo red removal via adsorption by sargassum fusiforme activated carbon could be well described by pseudo-second-order rate equation. The equilibrium removal rate and maximum adsorption capacity reach up to 94.72%, $234 \text{ mg} \cdot \text{g}^{-1}$, respectively when initial congo red concentration is $200 \text{ mg} \cdot \text{L}^{-1}$ under adsorbent dosage ($0.8 \text{ g} \cdot \text{L}^{-1}$), temperature (30°C), PH7.

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Supplementary Material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cjche.2020.01.016>.

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